



IOT PROJECT

TITLE:

KITCHEN INGREDIENTS SYSTEM

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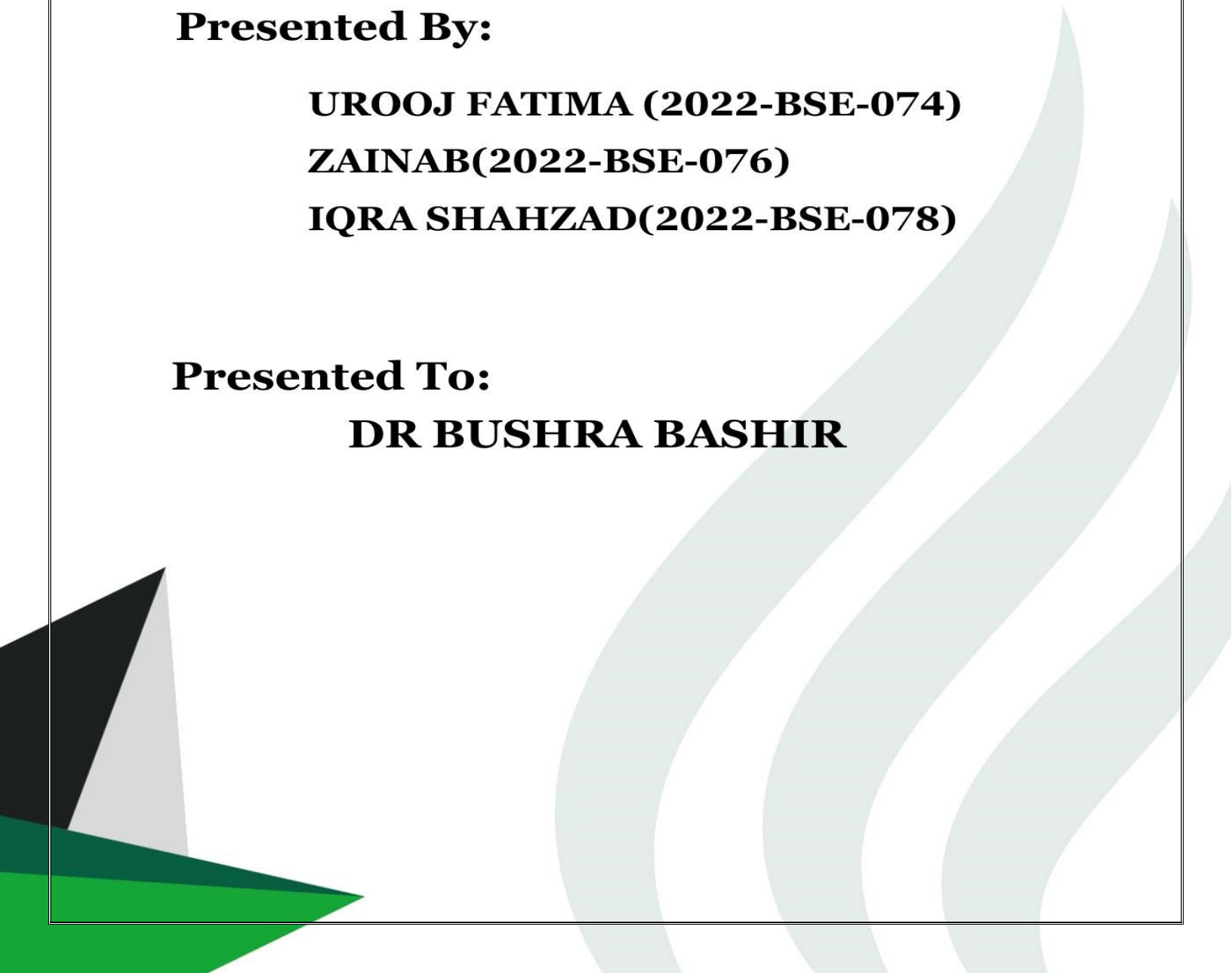


Table of Contents

Smart Kitchen Ingredient Monitoring System	3
Introduction	3
1. Initial Project Goal.....	3
2. Problem Statement.....	3
3. Problem Identification & Explanation	3
4. Feasibility Study.....	3
5. Alert Mechanisms.....	3
6. IoT System Overview	4
7. Survey Insights	4
8. Survey Outcomes	4
System Architecture.....	6
Virtual Hardware	6
Connections	6
Code.....	6
Physical Hardware Setup.....	8
Project Demonstration Video:.....	8
Wiring Diagram(Fritzing Diagram)	9
Blynk Setup	9
Dashboard(Blynk)	10
Cloud Database Setup.....	11
Email Notification	12
Future Work	12
Conclusion.....	12
References	12

Smart Kitchen Ingredient Monitoring System

Introduction

1. Initial Project Goal

The goal of this project is to design an **IoT-based smart kitchen ingredient monitoring system** that continuously monitors the quantity of sugar or any other item using a weight-based sensing mechanism. The system provides real-time feedback and alerts users when the ingredient level falls below a predefined threshold. The solution aims to reduce manual checking and improve kitchen inventory management.

2. Problem Statement

In many households, especially among students, working individuals, and busy families, essential kitchen ingredients often run out unexpectedly. Manual monitoring is inefficient and time-consuming, leading to inconvenience and last-minute shortages. There is a need for an **automated IoT solution** that can monitor ingredient levels and notify users in real time.

3. Problem Identification & Explanation

The problem is identified from real-life kitchen scenarios where users forget to check ingredient levels regularly. This IoT-based system offers a viable solution by continuously monitoring ingredient weight and generating alerts through visual, audio, and mobile notifications. The implementation is practical, reliable, and suitable for everyday use.

4. Feasibility Study

This project is technically and economically feasible for students. The required IoT components such as ESP32, sensors, LCD, and buzzer are low-cost and easily available. The system can be implemented at home using virtual hardware and cloud platforms, making it ideal for academic and practical learning.

5. Alert Mechanisms

The system provides low-ingredient alerts using multiple IoT-based notification methods:

- **LCD Display:** Displays messages such as “Low Ingredient” when sugar quantity drops below the threshold.
- **Buzzer Alert:** Produces an audible warning for immediate attention.
- **Mobile Notification:** Sends real-time alerts to the user’s smartphone using an IoT

cloud platform (Blynk).

6. IoT System Overview

The IoT system integrates sensors with an ESP32 microcontroller to collect data and transmit it to the cloud. The cloud platform stores the data and displays it on a mobile dashboard in real time. Users can monitor ingredient status remotely, making the system efficient and user-friendly.

7. Survey Insights

To validate the idea, a survey was conducted with 37 participants, including students and homemakers. When asked, “*Would you buy a small, low-cost ingredient tracking device?*”, **89.2% responded positively**.

- This result highlights strong demand for an IoT-based automated kitchen inventory monitoring system.

8. Survey Outcomes

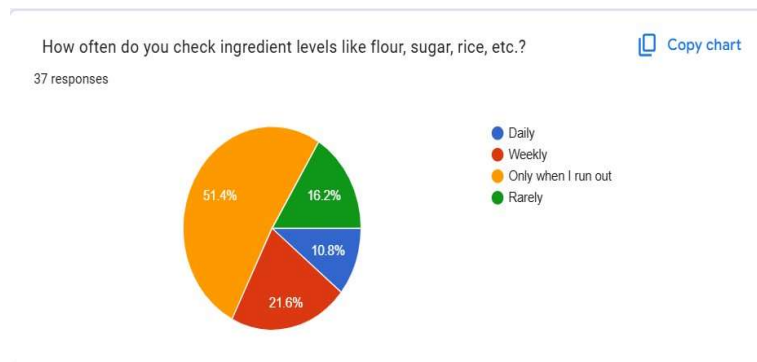


Figure 1: Ingredient Check Frequency: Most people (51.4%) check ingredient levels only when they run out.

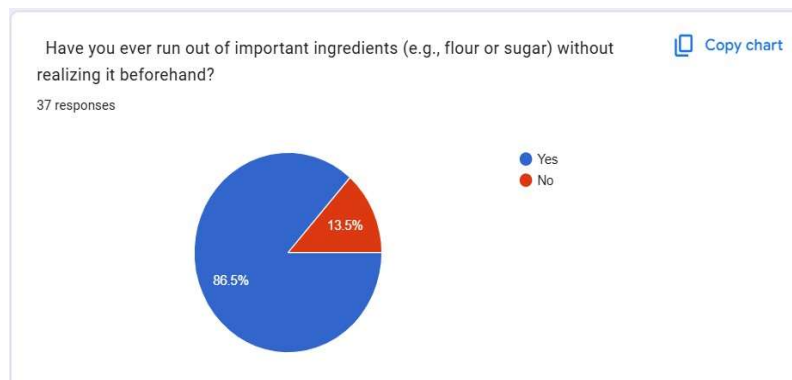


Figure 2: Running Out Unknowingly: 88.5% have experienced running out of important ingredients without realizing it.

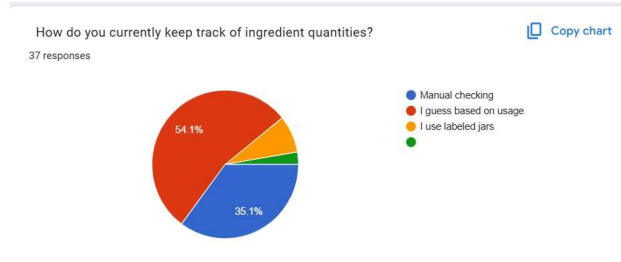


Figure 3: Tracking Method: 54.1% of people guess ingredient quantities based on usage rather than checking manually.

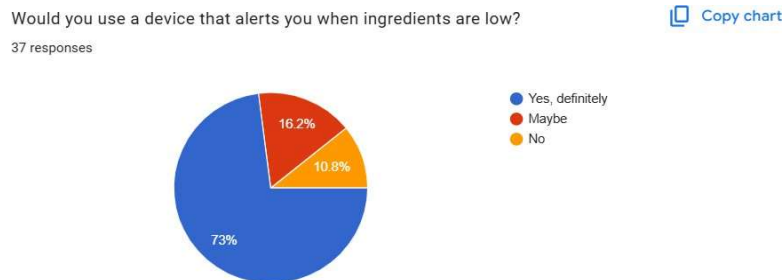


Figure 4: Device Usefulness: 73% of respondents would definitely use a device that alerts them when ingredients are low.

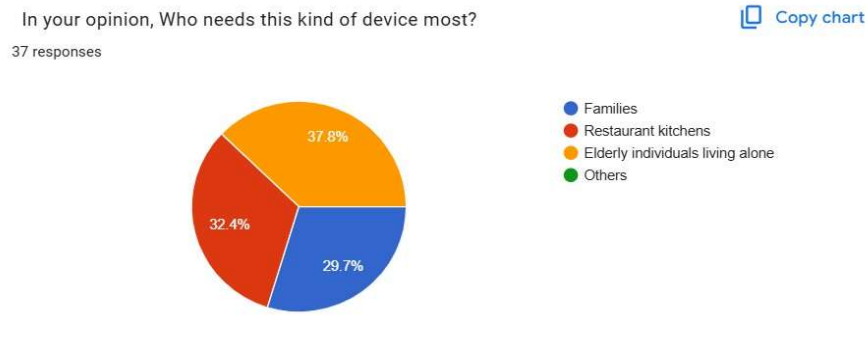


Figure 5: Target Users: Restaurant kitchens (37.8%) are seen as the group most in need of such a device

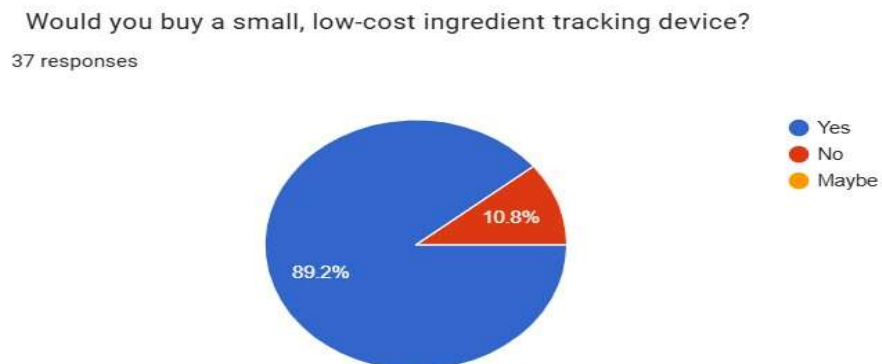
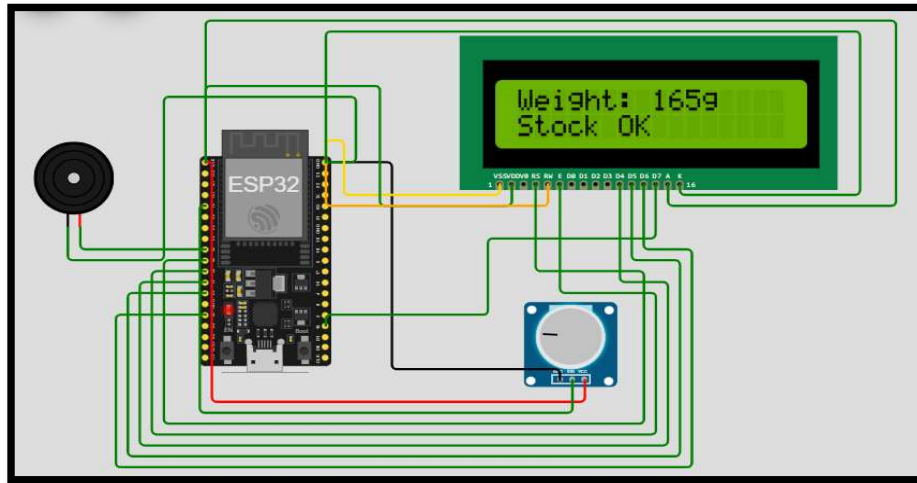


Figure 6: Willingness to Buy: A large majority (89.2%) would buy a small, low-cost ingredient tracking device.

System Architecture

Virtual Hardware



Connections

Component	Pin Name	Connected To (ESP32 Pin)	Purpose
ESP32	VIN / 5V	Power Source	Supplies power to components
ESP32	GND	Common Ground	Ground reference
Potentiometer	VCC	3.3V (ESP32)	Power supply
Potentiometer	GND	GND (ESP32)	Ground
Potentiometer	OUT	GPIO 34	Weight input (analog)
Buzzer	+	GPIO 25	Low stock alert
Buzzer	-	GND	Ground
LCD (16×2)	VSS	GND	Ground
LCD	VDD	5V	Power supply
LCD	V0	Potentiometer (contrast)	LCD contrast control
LCD	RS	GPIO 26	Register select
LCD	RW	GND	Write mode
LCD	E	GPIO 27	Enable signal
LCD	D4	GPIO 14	Data bit
LCD	D5	GPIO 12	Data bit
LCD	D6	GPIO 13	Data bit
LCD	D7	GPIO 15	Data bit

Code

```
#define BLYNK_TEMPLATE_ID "TMPL6PRyIDoxn"6
```

```
#define BLYNK_TEMPLATE_NAME "Smart Inventory Monitoring"
#define BLYNK_AUTH_TOKEN "AfhozSCnlmIzOML1aLkCzkMLWfKqseVa"
```

```
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <LiquidCrystal.h>
```

```
// ----- Hardware Pins -----
#define POT_PIN 34
#define BUZZER 25
#define RS 26
#define E 27
#define D4 14
#define D5 12
#define D6 13
#define D7 15
```

```
LiquidCrystal lcd(RS, E, D4, D5, D6, D7);
```

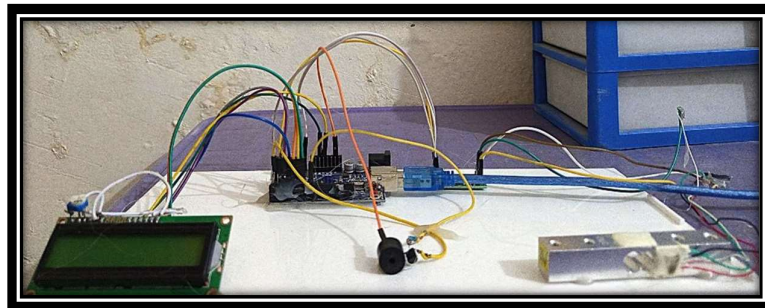
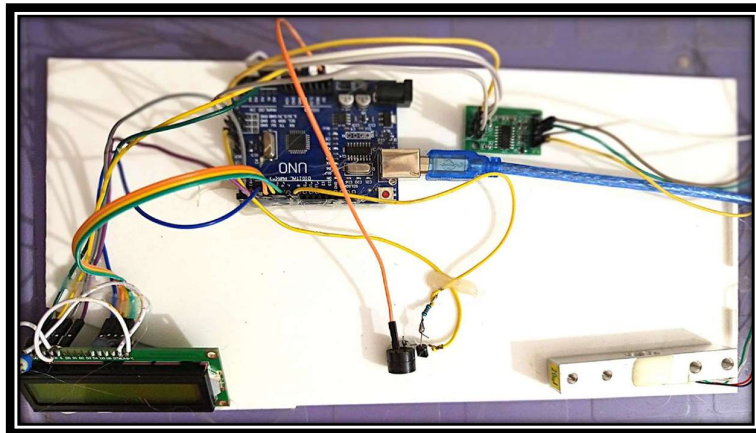
```
// ----- WiFi -----
char ssid[] = "Wokwi-GUEST";
char pass[] = "";
// ----- Variables -----
int rawValue = 0;
int weight = 0;
BlynkTimer timer;
// ----- SEND DATA TO BLYNK -----
void sendToBlynk() {
  // Gauge
  Blynk.virtualWrite(V0, weight);
  Blynk.virtualWrite(V3, "📦 SMART INVENTORY MONITORING");
  // Status + LED + Notification
  if (weight < 100) {
    Blynk.virtualWrite(V1, "LOW STOCK");
    Blynk.virtualWrite(V2, 1);    // LED ON
    Blynk.logEvent("low_stock", "Ingredient below 100g");
  } else {
    Blynk.virtualWrite(V1, "STOCK OK");
    Blynk.virtualWrite(V2, 0);    // LED OFF
  }
}
void setup() {
  Serial.begin(9600);
  pinMode(BUZZER, OUTPUT);
  // LCD
  lcd.begin(16, 2);
  lcd.print("Weight");
  lcd.setCursor(0, 1);
  lcd.print("Measurement");
  delay(1200);
  lcd.clear();
  // Blynk Connect
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  // Timer (same speed as loop delay)
  timer.setInterval(1000L, sendToBlynk);}
7
```

```

void loop() {
  // ----- Read Pot -----
  rawValue = analogRead(POT_PIN);
  weight = map(rawValue, 0, 4095, 200, 0);
  // ----- LCD -----
  lcd.setCursor(0, 0);
  lcd.print("Weight:  ");
  lcd.setCursor(8, 0);
  lcd.print(weight);
  lcd.print("g");
  // ----- BUZZER LOGIC -----
  if (weight < 100) {
    tone(BUZZER, 1000);
    lcd.setCursor(0, 1);
    lcd.print("Low Stock !!! ");
  } else {
    noTone(BUZZER);
    lcd.setCursor(0, 1);
    lcd.print("Stock OK  "); }
  Serial.print("Weight: ");
  Serial.println(weight);
  Blynk.run();
  timer.run();}

```

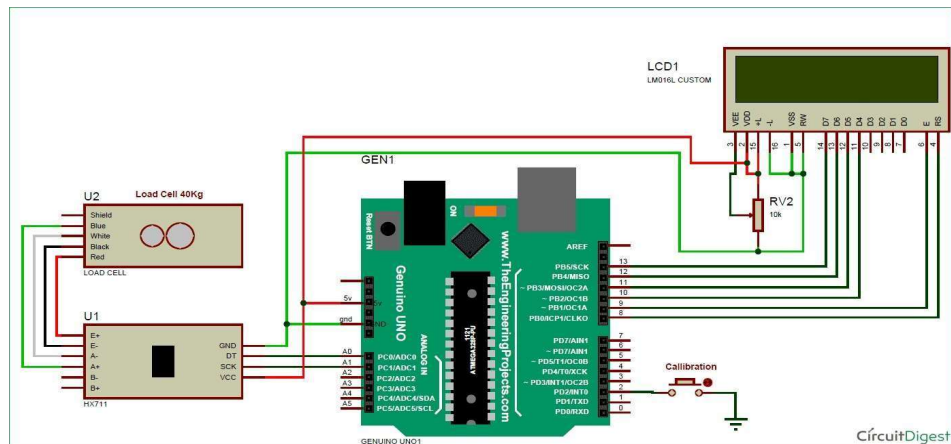
Physical Hardware Setup



Project Demonstration Video:

<https://drive.google.com/file/d/1Hat9LZs1pPW6q048xU98MdBhpKb0NCE2/view?usp=sharing>

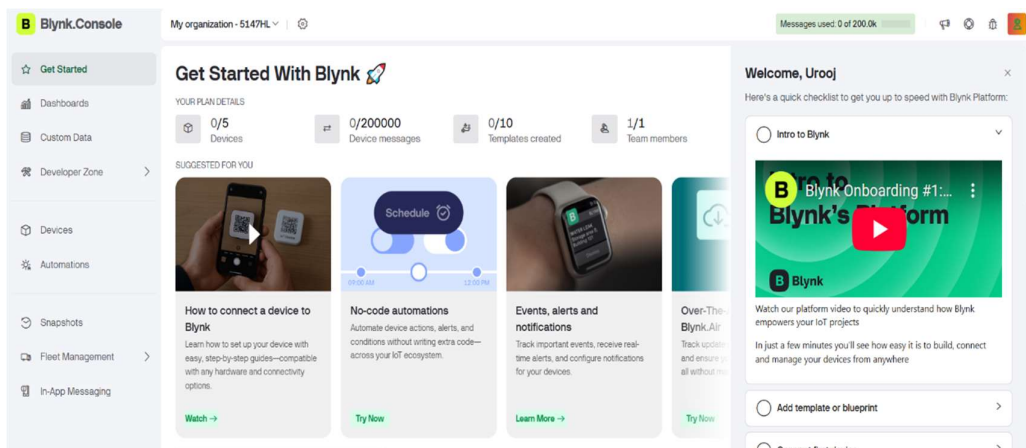
Wiring Diagram(Fritzing Diagram)



Blynk Setup

Blynk is used to provide a cloud-based frontend and database for the ESP32 system. It enables real-time monitoring of inventory weight and stock status through a mobile/web dashboard.

The ESP32 communicates with Blynk over Wi-Fi, allowing remote data visualization, alerts, and notifications.



Create New Template

NAME
Smart Inventory Monitoring 26 / 50

HARDWARE
ESP32

CONNECTION TYPE
WiFi

DESCRIPTION
Description 0 / 128

Cancel Done

Virtual Pin Datastream

Weight

PIN: V0 DATA TYPE: Integer

UNITS: Gram, g

MIN: 0 MAX: 500 DEFAULT VALUE: 0

☒ Enable history data

ADVANCED SETTINGS

Cancel Create

Smart Inventory Monitoring

Cancel Save

Datastreams

Search datastream + New Datastream

2 Datastreams

ID	Name	Pin	Color	Data Type	Unit	Actions
1	Weight	V0	Orange	Integer	g	
2	Stock Status	V1	Red	String		

New Device

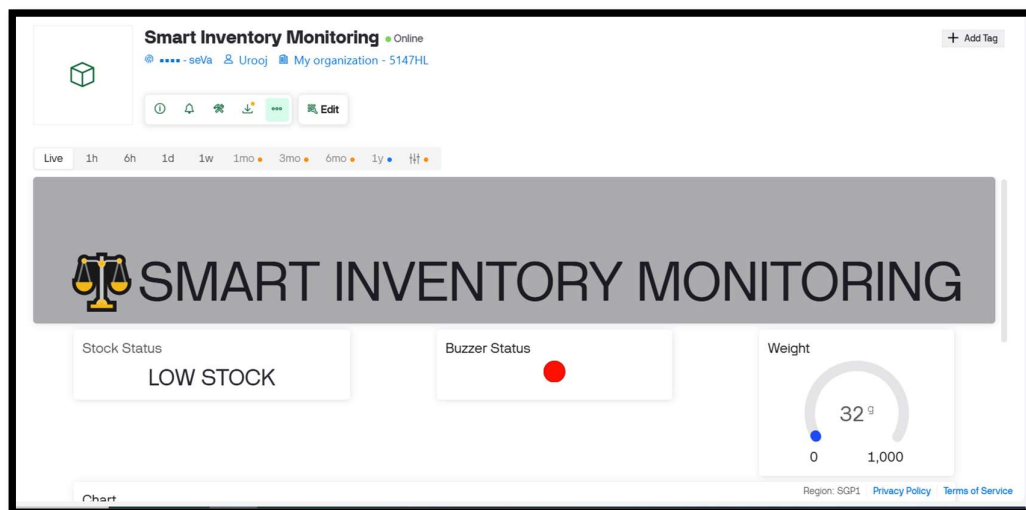
Create new device by filling in the form below

TEMPLATE: Smart Inventory Monitoring

DEVICE NAME: Smart Inventory Monitoring 26 / 50

Cancel Create

Dashboard(Blynk)





Results & Discussions

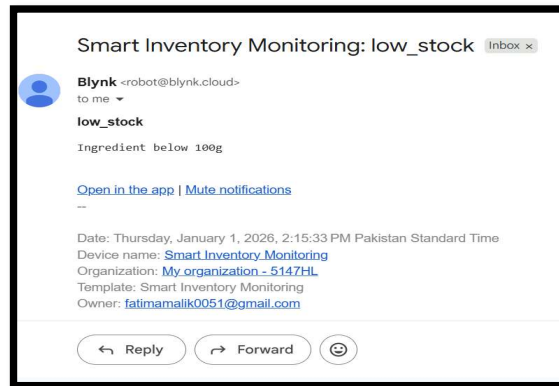
The Blynk dashboard successfully displays real-time and historical weight data received from the ESP32. The line chart shows continuous variations in stock level over time, while the bar-type chart highlights sudden changes in weight, indicating stock consumption events. During the low-stock condition, the gauge displays a value of **32 g**, the status label changes to “**LOW STOCK**”, and the LED turns **red**, indicating an alert state. Simultaneously, the buzzer is activated, confirming that the system accurately detects low inventory levels.

Cloud Database Setup

The figure shows the Blynk Cloud Database Setup interface. The top part is the 'Events & Notifications' section, which includes a 'Create Event' button. The bottom part is the 'Add New Event' form, which has tabs for 'General', 'Notifications', 'Settings', and 'Expose to Automations'. The 'General' tab is active, showing fields for 'EVENT NAME' (low_stock), 'EVENT CODE' (low_stock), 'TYPE' (Warning), and 'DESCRIPTION (OPTIONAL)' (Ingredient level below threshold Δ).

Email Notification

- The system alerts the user about low_stock.



Future Work

Future enhancements to further improve the system could include:

- Implementation of **AI-based demand prediction** to forecast inventory requirements and optimize stock levels.
- Adding **automatic reordering** when stock falls below a predefined level.
- Use of **cloud analytics** to track trends and generate historical reports automatically.
- Integration with **IoT-enabled smart shelves** for automated inventory scanning.

Conclusion

This project successfully implemented a smart inventory monitoring system using ESP32 and Blynk IoT platform. The system accurately monitors stock weight in real time and provides instant visual and audible alerts during low-stock conditions. Reliable cloud connectivity ensures continuous data logging, making the system efficient, user-friendly, and suitable for real-world inventory management applications.

References

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